The Case For A Local GPU Workstation

A 15 000 EUR investment that pays for itself in months

Raul Llamas-Sandin, Universidad de Sevilla

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The Ask, In One Slide

Request

Approval for a **single workstation purchase up to 15 000 EUR** (Spain, all-in: GPU, CPU, RAM, storage, chassis, support).

Two reasons, both already proven on our current small GPU:

1. **Numerical solvers** – our GPU-native CFD and topology codes are bottlenecked by VRAM and SM count, not by our algorithms.
2. **Local AI inference** – runs the AI agents that produced most of our software output in the last year, with privacy, no per-token cost, and no internet dependence.

The same hardware serves both jobs.

Proof Of Value – Current Results

On the existing modest GPU we have:

- Brought **OpenLUDWIG** (compressible 3D LBM CFD) from zero to running real cases. Performance is limited by VRAM today – domains are smaller than they need to be.
- Run **HEXATOP** (3D topology optimization, PhD core) on representative bracket and panel problems.
- Run **OpenFLIGHT** as an interactive real-time simulator with full non-linear aero data.
- Used **local LLM inference** for routine code edits and sensitive content, with cloud frontier models reserved for hard planning.

Illustration to insert:

Screenshot: OpenLUDWIG running a wing-body configuration, with VRAM usage indicator at ∼95 % on the current GPU.

What The Small GPU Cannot Do Today

- Run OpenLUDWIG on **full-aircraft** domains at the resolution we need for design-quality loads.
- Run HEXATOP on **industrial-scale** optimization domains (millions of design variables).
- Host a serious local LLM at **useful context lengths** (we are limited to small models with short contexts).
- Multi-tenant: cannot run a solver and an LLM at the same time.
- Drive any meaningful **fine-tuning** of a local model on our codebase.

These are exactly the workloads we want next.

When GPU Solvers Make Sense

GPUs are not always the right answer. They shine when:

- the algorithm is **massively parallel and local** (LBM, explicit time integration, topology optimization with cellular updates);
- the per-step memory footprint fits in VRAM;
- bandwidth, not flops, is the bottleneck;
- iteration count is large enough to amortize host–device transfer.

They are wrong for:

- small implicit problems with heavy sparse factorization,
- workflows dominated by data movement to and from disk,
- one-shot computations that finish in seconds on CPU.

Commercial CFD With GPU Back-Ends Today

Where the industry stands as of late 2026:

- **Ansys Fluent** – full GPU solver paths for steady and transient incompressible / mildly compressible work.
- **Siemens StarCCM+** – GPU acceleration for selected solver paths.
- **Cadence Fidelity** – end-to-end GPU CFD product line.
- **Altair AcuSolve / ultraFluidX** – GPU LBM offering for external aerodynamics.
- **OpenFOAM** – third-party GPU back-ends (PETSc-based, AmgX), production-grade but partial.

OpenLUDWIG sits in the LBM segment, alongside ultraFluidX. The market validates the bet.

Solution Time Comparison

Illustration to insert:

Bar chart: representative wall-clock for one external-aerodynamics CFD case across (a) CPU OpenFOAM, (b) GPU OpenLUDWIG on current small card, (c) projected GPU OpenLUDWIG on the proposed workstation. Mark VRAM ceiling on each.

Order-of-magnitude expectation: moving from 12–16 GB VRAM today to $\geq$48 GB on the proposed hardware unlocks domain sizes we cannot touch at present.

Why Julia Makes The GPU Bet Safer

- One language for host code *and* GPU kernels via CUDA.jl, AMDGPU.jl, or oneAPI.jl.
- No CMake / Cython / pybind11 layer to maintain.
- Kernels are generic functions: the same code can target NVIDIA, AMD, Intel, or CPU fallback.
- Composable AD (Enzyme, Zygote, ForwardDiff) extends to GPU code, enabling adjoint topology and shape optimization.

Practical consequence: a GPU rewrite that costs a quarter in C++/CUDA costs weeks in Julia, and stays portable.

The Case For Local AI

Why run inference (and selective training) locally:

- **Privacy** – sensitive code, customer data, draft IP never leaves the building.
- **Cost predictability** – a fixed hardware cost replaces variable per-token charges that scale with our productivity.
- **Latency** – no network round trip, fast iteration.
- **Offline** – the team is productive without internet.
- **Fine-tuning** on our own codebase – impossible at our scale on cloud-frontier models, feasible on a local open-weights model.

Cloud frontier models remain useful for the hardest planning. The workstation makes that the exception, not the default.

Local Models Worth Running In Late 2026

Family	Notes	Use case	Min VRAM (quant.)
Gemma 3 (Google)	Multiple sizes, strong reasoning per parameter	Routine code, doc generation	12–48 GB
Qwen 3	Excellent code performance, multilingual	Code edits, refactors	16–64 GB
DeepSeek-R1 distills	Reasoning-focused, smaller distilled variants	Hard planning offline	24–48 GB
Llama 3.x	Broad ecosystem, many fine-tunes	General assistant	16–48 GB
Mistral / Codestral	Lightweight, fast on consumer cards	Linting, refactor	8–24 GB

A 48 GB card lets us run a strong code model at useful context, or two smaller models in parallel. A 96 GB card opens fine-tuning at full precision and very long contexts.

GPU Options Under 15 000 EUR (Spain, Late 2026)

GPU	VRAM	Best for	Price band (approx.)
NVIDIA RTX 5090	32 GB	Single-GPU sweet spot; OpenLUDWIG mid-domain; LLM up to $\sim$30 B	2 500–3 000 EUR
2$\times$ NVIDIA RTX 5090	64 GB total	Multi-GPU LBM; 70 B-class LLM via tensor parallel	5 500–6 500 EUR
NVIDIA RTX PRO 6000 Blackwell	96 GB	Single-card workstation king; large CFD domains; long-context LLM; fine-tuning	8 500–10 500 EUR
NVIDIA H100 (PCIe, used / refurb)	80 GB	Data-center grade, ECC, NVLink in pairs; the inference benchmark to beat	12 000–15 000 EUR

Recommended primary path: **single RTX PRO 6000 Blackwell (96 GB)** – maximum VRAM per slot, professional drivers, ECC, fits one chassis, leaves budget for the rest of the machine.

Proposed Build (Recommendation)

Component	**Spec**	**Approx. EUR**
GPU	NVIDIA RTX PRO 6000 Blackwell, 96 GB GDDR7 ECC	9 500
CPU	AMD Threadripper 7965WX (24 cores, 5 GHz boost)	1 800
Motherboard	WRX90 workstation board, 8 DIMM, 7 PCIe 5.0 slots	900
RAM	256 GB DDR5-5600 ECC RDIMM (8×32 GB)	1 400
Storage	2 TB Gen5 NVMe (OS) + 8 TB Gen4 NVMe (data)	700
Chassis + PSU	Workstation tower, 1600 W Platinum	500
Misc + 3y on-site	Cables, fans, vendor support	200
Total		**≈15 000**

Prices are indicative for the Spanish market, late 2026, business channel. Exact configuration to be confirmed at purchase time.

Alternative Build (Multi-GPU Path)

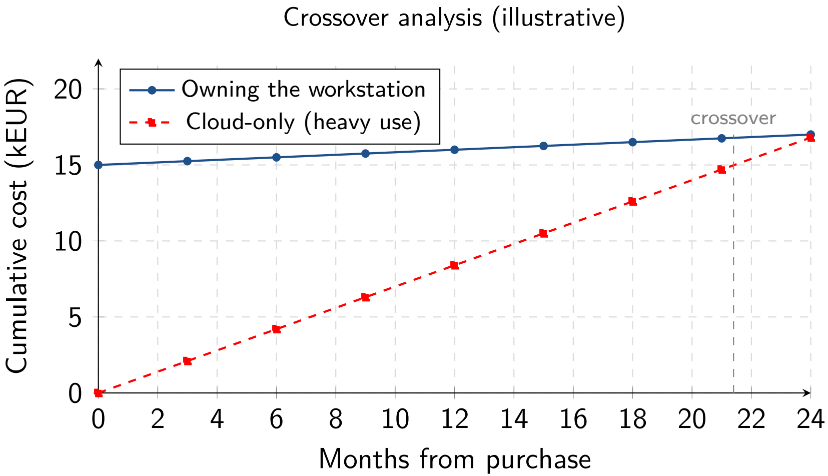
Component	**Spec**	**Approx. EUR**
GPU	$2\times$ NVIDIA RTX 5090 (32 GB each)	6 000
CPU	AMD Threadripper 7975WX (32 cores)	2 800
Motherboard	WRX90 workstation board	900
RAM	256 GB DDR5-5600 ECC RDIMM	1 400
Storage	2 TB Gen5 NVMe + 8 TB Gen4 NVMe	700
Chassis + PSU	Workstation tower, 1600 W Platinum	500
Misc + 3y on-site	Cables, fans, vendor support	200
Contingency		1 500
Total		**≈14 000**

Pros: higher total flops, MIG-like flexibility (one GPU per workload). Cons: 64 GB pooled VRAM only via tensor parallelism, two cards is more cooling, and the 5090 has no ECC.

Why The Single-Card Path Wins For Us

- Our biggest pain is **not flops, it is VRAM**. 96 GB is the largest single number we can buy in this price band.
- **ECC memory** matters for long CFD runs and overnight training.
- One card means one solver *and* one LLM coexist without contention.
- Professional drivers and 3-year on-site warranty – the machine has to work every day.
- Future-proof for two years of model growth.

Cloud Inference vs. Owning The Hardware



Even on a conservative cloud spend, the hardware crosses over inside two years. With current usage trends it crosses much sooner. Numbers illustrative; real figures depend on which cloud tier we use.

New Capabilities Unlocked

- Run OpenLUDWIG on **full-aircraft** loadcases overnight (current limit: small panels).
- Run HEXATOP on **industrial-scale** optimization domains with millions of design variables.
- Host a serious local LLM (70 B-class or a strong 30 B at long context) for sensitive code work.
- Fine-tune a local model on our own codebase and reports – producing an in-house assistant that "knows" our solvers.
- Run solver and LLM **concurrently** without contention.

Risk And Mitigation

Risk	Mitigation
GPU prices fall after purchase	The card is bought for capability, not speculation. Useful for at least 4 years given workload growth.
A better model is released in 6 months	Open-weights models are getting smaller for the same capability; same VRAM keeps working.
Cloud becomes much cheaper	We continue to use cloud for the hardest planning; the workstation absorbs routine and sensitive work.
GPU fails / DOA	3-year on-site warranty, Spanish business channel.
Software stack changes	Julia + standard CUDA stack are well supported; no exotic dependencies.

What We Need From You

1. **Approval** of the 15 000 EUR budget for one workstation.
2. **Choice** between the single-card (RTX PRO 6000 Blackwell) and dual-card (2× RTX 5090) configurations.
3. Three-year **on-site warranty** included.

Once approved, lead time in Spain for either configuration is typically 2–4 weeks.

We can demonstrate every claim in this deck on the current small GPU before the purchase is finalized.

Closing

We have already shown what is possible on **modest hardware and a disciplined workspace**.

A 15 000 EUR workstation removes the only constraint we still hit every day – VRAM – and at the same time hosts the AI infrastructure that lets a small team deliver like a large one.

It is the highest-leverage single purchase we can make right now.

Thank you. Ready for questions.